StockLens — Platform & Interface Overview

A comprehensive architectural and functional walkthrough of the four primary operational interfaces with side-by-side screenshots, functional breakdowns, and underlying quantitative mechanics.

DISCOVERY & PAIRING CONTROL CENTER · TAB 01

Route: /

System Online & Verified

Home Interface

The Home Tab serves as the central command portal for StockLens. It allows evaluators and students to immediately configure comparative equity evaluations through one-click curated pairs or arbitrary ticker exploration.

Core Capabilities & Operational Features:

- **9 Curated Exhibition Pairs:** One-click comparative analysis for leading sector peers (Apple vs Microsoft, Nvidia vs AMD, Coca-Cola vs PepsiCo, Tesla vs Ford, TVS vs Tata Motors, HAL vs Mazagon Dock).
- **Rapid Autocomplete Search:** Instant ticker and corporate name resolution against US & Indian equity databases with input validation.
- **Offline Exhibition Snapshot Mode:** Deterministic offline demo mode (Ctrl+Shift+D) backed by pre-fetched, audited datasets for guaranteed zero-latency demos without network reliance.
- **Mobile Handoff QR Code:** On-screen QR generator enabling judges and exhibition visitors to scan and open the active comparison on their personal mobile devices.

Operational Workflow: Select two companies via autocomplete or preset pill -> Preview summary parameters -> Launch deep-dive evaluation via the primary 'Run Comparative Analysis' action.

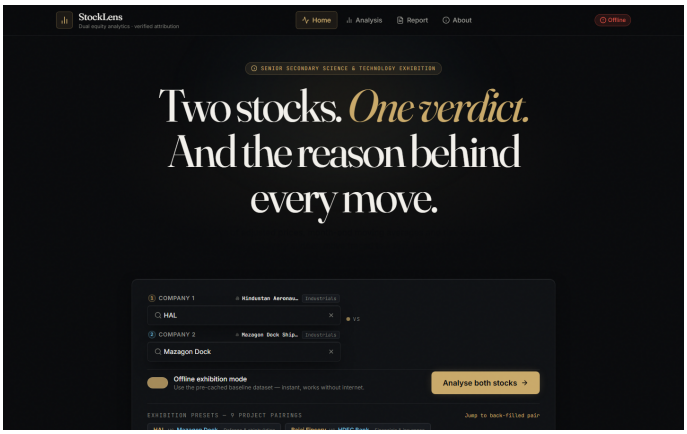


Figure 1: StockLens Home Interface — Dual company selector, curated exhibition pairs, and offline snapshot switcher.

Technical Architecture & Implementation Details:

Architecture & Stack: React 18, TypeScript, Tailwind CSS · **Key Modules:** Home.tsx, CompanyInput.tsx · **State Management:** Active pairing cached in browser localStorage for persistent cross-tab routing · **Modes:** Normal, Kiosk (?kiosk=1, 90s reset), Presenter Mode.

DEEP QUANTITATIVE COMPARATIVE ENGINE · TAB 02

Route: /analysis

System Online & Verified

Analysis Interface

The flagship quantitative dashboard delivering side-by-side evaluation across 2-year adjusted closing prices. It marries mathematical risk and return metrics with verifiable event attribution and algorithmic decision scoring.

Core Capabilities & Operational Features:

- **Synchronized Interactive Visualizations:** Normalized performance charts (base 100), 3-month rolling moving averages, historical drawdown profiles, and rolling volatility curves.
- **Mathematical Risk/Return Statistics:** Rigorous calculation of Cumulative Return, CAGR, Annualized Volatility (ddof=1, √252), Sharpe-style ratio, Beta, and Pearson Correlation.
- **'Why It Moved' Anomaly Engine:** Flags statistically anomalous daily price moves exceeding 2.5 standard deviations (σ) and links contextual Google News headlines with publisher attribution.
- **Deterministic Verdict Scorecard:** Six weighted rule-based signals (Trend, Momentum, Volatility, Sharpe Quality, 52-Week Position, Valuation Proxy) mapped into clear Buy, Hold, or Avoid recommendations.

Operational Workflow: Review KPI cards -> Evaluate rebased growth curves -> Inspect flagged historical anomaly dates -> Review signal breakdown behind algorithmic verdict.

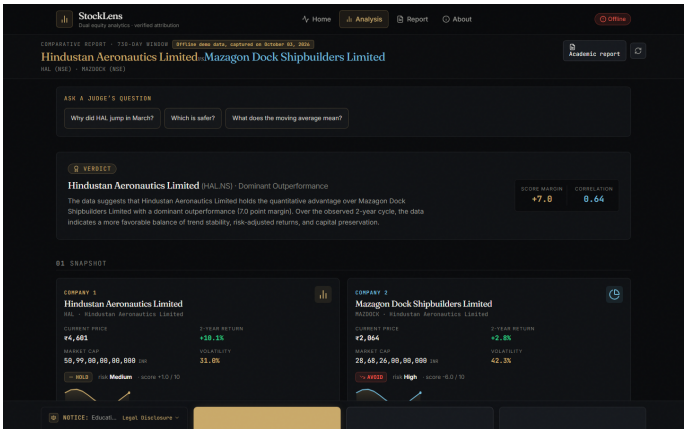


Figure 2: StockLens Analysis Dashboard — Synchronized interactive charts, volatility bands, and rule-based verdicts.

Technical Architecture & Implementation Details:

Architecture & Stack: Recharts, Lucide Icons, FastAPI · **Key Modules:** Analysis.tsx, TerminalChart.tsx, MetricCard.tsx · **Hotkeys:** Press 'P' for auto-scrolling Presenter Mode · **Data Flow:** Yahoo Finance adjusted price pipeline with statistical anomaly filtering.

EXECUTIVE AUDIT DOSSIER & EXPORT HUB · TAB 03

Route: /report

System Online & Verified

Report Interface

The Report Tab converts dual-equity mathematical evaluations into an executive-ready, printable dossier. Crafted with print-first styling and high information density, it is structured for physical printing, PDF distribution, and formal judge assessment.

Core Capabilities & Operational Features:

- **Executive Summary Matrix:** Clear side-by-side scorecard highlighting performance leaders, risk minimizers, and Sharpe winners across all evaluated metrics.
- **Month-End Tabular Breakdown:** Full historical table of monthly adjusted close prices, moving averages, and monthly percentage changes with sorted alignment.
- **Chronological Anomaly Log:** Complete audit trail of flagged unusual volatility days, complete with dates, percentage price moves, and corroborated news source links.
- **Browser & PDF Print Layouts:** Clean CSS media print styles that eliminate extraneous UI controls, banners, and interactive buttons for pristine paper or PDF output.

Operational Workflow: Review executive summary comparisons -> Filter monthly pricing table -> Export clean CSV for spreadsheet modeling -> Trigger 1-click Print/PDF export for paper presentation.

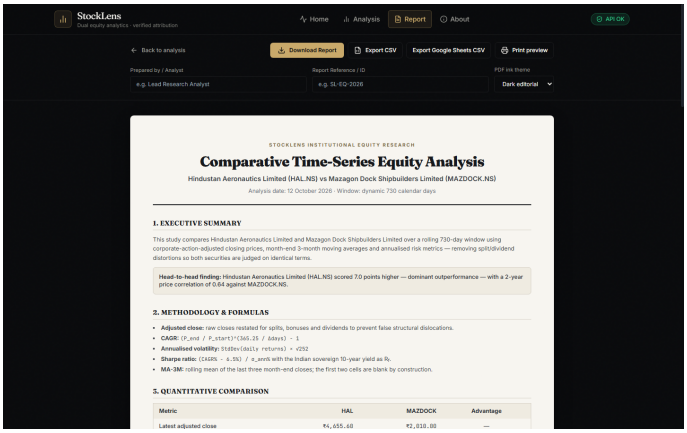


Figure 3: StockLens Executive Report — Print-first layout with high-density metric matrices and compliance audit logs.

Technical Architecture & Implementation Details:

Architecture & Stack: CSS Paged Media, Vite · **Key Modules:** Report.tsx, SummaryMatrix.tsx, PrintHeader.tsx · **Export Formats:** Direct PDF via browser print engine, formatted CSV download · **Compliance:** Embedded timestamp and legal disclaimer stamps.

METHODOLOGY, MATHEMATICS & GOVERNANCE · TAB 04

Route: /about

System Online & Verified

About Interface

The About Tab provides complete transparency into the quantitative logic, statistical parameters, attribution boundaries, and academic principles underlying StockLens. It reinforces the system's commitment to verifiable and explainable analytics.

Core Capabilities & Operational Features:

- **Mathematical Formula Transparency:** Explicit documentation of all equations including CAGR: (last/first)^(365.25/days)-1, Volatility: s*√252, and Sharpe-style metric: (CAGR - Rf)/Volatility.
- **Attribution Governance & Ethics:** Clear delineation between observed correlation and causal claims: historical news headlines provide contextual evidence rather than causal proof.
- **Signal Weighting Disclosure:** Transparent inspection of the six signal weights powering the recommendation engine, ensuring zero black-box obscurity.
- **Educational & Regulatory Disclaimer:** Explicit disclaimers establishing that StockLens is an educational demonstration project and does not offer financial or investment advice.

Operational Workflow: Read five-step explainer -> Inspect mathematical formulas behind risk metrics -> Verify signal scoring weights -> Review judge Q&A; and attribution guidelines.

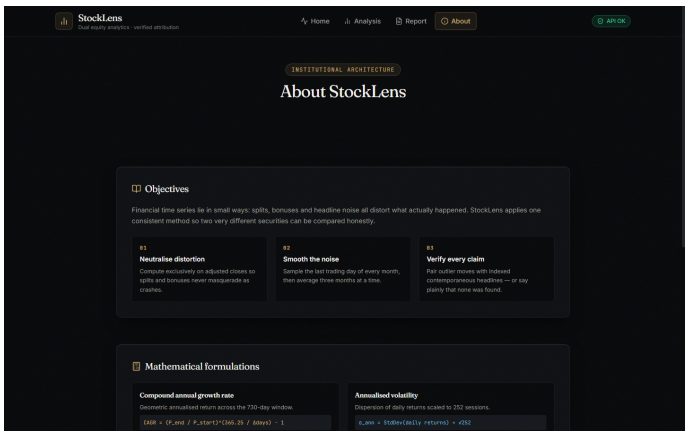


Figure 4: StockLens About & Methodology — Explicit mathematical definitions, statistical thresholds, and ethical safeguards.

Technical Architecture & Implementation Details:

Architecture & Stack: Markdown documentation, Lucide Icons · **Key Modules:** About.tsx, FormulaCard.tsx · **Integrity Safeguards:** Data cleaning rules for dividends/splits, missing date handling · **Pedigree:** Developed for senior secondary science & technology exhibition.